

U-Net Based Fetal R-peak Prediction From Abdominal ECG Signals

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Abstract—Fetal heart rate (FHR) monitoring is an important part of antenatal care, providing important indicators of fetal health and diagnosis of fetal disease. Non-invasive fetal electrocardiograms (NI-FECG) for FHR monitoring has received much attention due to its suitability for continuous monitoring, unlike other currently used technologies. However, NI-FECG utilises an abdominally recorded ECG (AECG), which is a mix of maternal and fetal ECG along with other noise signals. Due to the low SNR of the FECG signal, extraction of the FECG to determine FHR is a challenging task. In recent years, various deep learning based methods have shown promising results in predicting fetal R-peak locations used to find the FHR. Most of these methods begin by predicting and cancelling the maternal ECG (MECG) signal prior to a second fetal R-peak detection process.

This work proposes an R-peak detection method based on the U-Net architecture for direct detection of fetal R-peaks from the AECG without the need for prior cancelling of the MECG. The proposed method includes a deep learning model based on a shallow U-Net structure, combined with Bidirectional Long-Short Term Memory (Bi-LSTM) and residual paths. Performance of the proposed detection method is evaluated on AECG recordings from set-A of the PCDB PhysioNet database. The proposed method was compared with five other approaches which perform direct fetal R-peak detection using machine or deep learning. Numerical results demonstrate that the proposed UNet-BiLSTM model has reliable performance for fetal R-peak detection from AECG signals, reaching 93.73%, 91.10% and 92.35% in average PPV, Recall and F1-scores, respectively, outperforming the other methods evaluated.

Index Terms—U-Net, LSTM, R-Peak Detection, Fetal Heart Rate Monitoring, Fetal ECG

I. INTRODUCTION

Fetal heart rate (fHR) is an important indicator crucial for assessing fetal health and diagnosing fetal disease. By utilizing the relatively high amplitude R peak in the ECG signal, the cardiac cycle can be identified and the heart rate accurately measured. However, since the non-invasive fetal (NI-FECG) ECG signal is affected by multiple sources of nonlinear noise including the maternal ECG signal which overlaps in both the time and frequency domains, it has a low signal-to-noise ratio (SNR) making accurate fetal R peak detection results difficult to realize.

In recent years, deep learning-based networks have achieved state-of-the-art performance in medical image segmentation, extracting key features from samples and determining optimal model parameters without any domain knowledge. This lends

to the viability of detecting fetal R-peaks directly from the abdominal ECG (AECG). While a considerable body of work in the processing of FECG from AECGs appears in the literature, few directly detect R-peaks from the AECG without significant preprocessing and separation of maternal from fetal ECG prior to detection.

Zhong et al. [1] were the first to propose using a CNN-based deep learning model to achieve direct detection of the fetal R-peak. Their model contains three volume-based layers, using a single-channel NI-FECG signal as input. Signal quality was evaluated via Sample Entropy calculations and then the single-channel NI-FECG signal was used as input to the model. Note that since this model only contains three volume base layers, it is not enough to accurately extract the characteristics of fetal QRS complexes in NI-FECG. Lee et al. [2] proposed a modified CNN model based on previous work by Zhong et al., adding a post-processing scheme which helped to enhance the fetal QRS in NI-FECG signals. Their method used multi-channel NI-FECG signals as input and a deeper architecture to fully extract the features of fetal QRS complexes in NI-FECG signals, resulting in improved R-peak detection accuracy.

The U-Net structure is a deep learning network that has shown powerful capabilities on medical image segmentation tasks. The architecture consists of contracting paths that capture context and symmetric expanding paths that enable precise positioning. Thus the U-Net's residual connections in an encoder-decoder structure enables it to be able to achieve detailed and accurate segmentation using limited samples for training [3]. Its context-based learning also enables U-net train much faster with fewer labelled samples than most other segmentation models [4]. Since U-Net based structure can be trained end-to-end to provide precise localization, it has been applied to a range of other applications. For example, U-Net was adapted for R peak detection in adult ECG signals [5]–[7], and blood pressure monitoring for photoplethysmography signals [8]. He et al. [6] modified the U-Net architecture to include a bidirectional long short term memory (Bi-LSTM) in the bottleneck of their architecture on two channel adult ECG signals, and showed improved performance compared to the traditional U-Net structure. This enabled important information contained in a longer time series to be retained, enabling it to extract contextual information and bidirectional

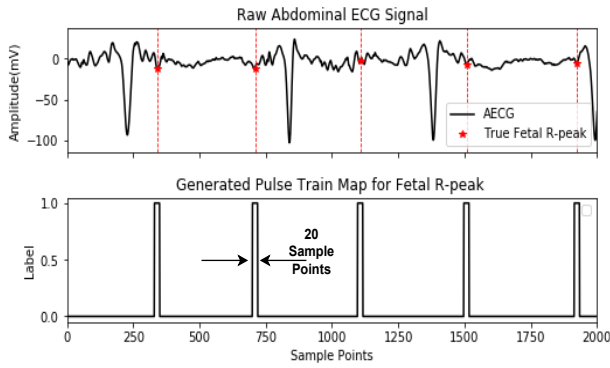


Fig. 1. An example of the Abdominal ECG signal and corresponding pulse train map based on fetal heart beats generated. The red star represents the true R-peak annotation of the fetal, corresponding to the center of each pulse in the lower image.

rhythm features of the ECG.

Motivated by these works, this paper proposes a model based on U-Net with Bi-LSTM blocks and residual connections for one-dimensional fetal R-peak detection directly from AECG. The proposed model was evaluated on the PhysioNet Cardiology Challenge Database (PCDB) from Physionet [9]. In spite of using a shallow U-Net structure (compared to previous works using U-Net) to reduce labelled data requirements, incorporation of the Bi-LSTM enabled improved performance compared with existing R-peak detection methods evaluated.

The remainder of this paper is organized as follows: Section II describes the proposed fetal R peak detection method. Section III provides details of the experiments performed, while section IV presents and discusses the experimental results. Finally, conclusions are drawn in section V.

II. METHODOLOGY

A. Data Segmentation and Labeling

For this work, a pulse train was first generated to represent the true annotations of each fetal R-peak in the dataset. Figure 1 shows an example segment from the original AECG signal with true fetal R-peak annotations highlighted, and the corresponding pulse train map generated. Here we use a pulse width of 20 sampling points, with 10 sample points on the left side and 9 sample points on the right of true R-peak location. Samples points for each pulse are labeled as one and remaining points are labeled as zero. The AECG signal and pulse train map of fetal R-peaks are then segmented with a length of 1000 sample points (1 second at 1000 Hz) and a shift of 200 sample points. This provides sufficient segments and information in each segment for model learning without excessive calculation cost.

B. Model Architecture

The proposed Unet-BiLSTM model represents an enhancement of the Vanilla U-Net architecture. The conventional U-Net structure comprises a contracting path(encoder), an expansive path(decoder), and a bottleneck layer that acts as a

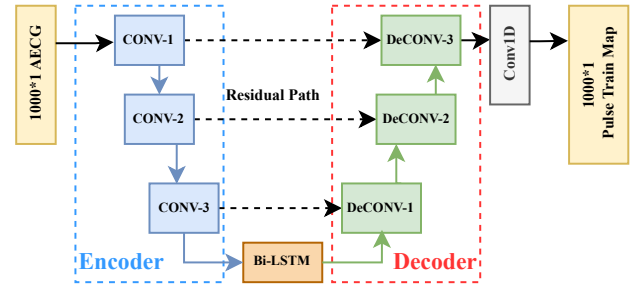


Fig. 2. Proposed Unet-BiLSTM Structure.

pivotal transition point between the contracting and expansive paths. In our proposed model, we introduce cascaded Bi-LSTM blocks to replace the conventional convolutional layers and residual paths within the bottleneck. The architecture of proposed network is illustrated in Figure 2.

The encoder section comprises three CONV-blocks. The filter parameters and kernel size for each block are 16, 16, 32 and 9,9,6 respectively. Each of the CONV-block has a one-dimensional convolutional layer, followed by a ReLU activation layer. A max-pooling layer with a size of 2 is then used to achieve down-sampling of the spatial dimensions. The bottleneck section is a Bi-LSTM block concatenated with three Bi-LSTM layers with 32 units for each. Each Bi-LSTM layer consists of one forward LSTM layer and one backward LSTM layer, facilitating the retention of important contextual and bidirectional rhythm and morphology information. Similarly, the decoder section consists of three DeCONV-blocks. Each of them include one deconvolutional layer, where the output of the deconvolutional layer is concatenated with the corresponding information transferred from the encoder via a residual path. These connections enable the network to preserve detailed spatial information from earlier layers while combining it with context learned from deeper layers. This facilitates precise positioning during segmentation tasks. A one-dimensional convolutional layer is then followed by a ReLU activation layer. The filter parameters and kernel sizes for each block are 32, 16, 16 and 9, 9, 6, respectively. Finally, a one dimensional convolutional layer is employed as the output layer with a Sigmoid activation function.

C. Fetal R-peak Detection Method

The proposed fetal R-peak detection applies the described segmentation to test AECG records to the model constructed in the training stage. The output of the model is a pulse train indicating predicted locations of R-peaks. Post processing steps are then applied to identify and eliminate false positives, thereby improving the accuracy of the R-peak detection method.

Firstly, the pulse train indicating predicting fetal R-peaks for each signal segment is converted into a full-length map using overlapping-add. In the second step, a threshold $THRES_{small}$ is applied, converting all predicted sample points with a value less than that threshold to zero. This is done to mitigate false positive predictions associated with weak predictions. In the

third step, the centre of each pulse is located in order to generate predicted fetal R-peak locations. This is followed by the application of a second threshold $THRES_{close}$ which eliminates predicted beats that are too close to be physically possible. In the case where two predicted beats are closer than this threshold, the one further from the midpoint of preceding and subsequent beats is removed.

The final result is a pulse train such that each pulse corresponds to the predicted location of a fetal ECG R-peak within the original AECG.

III. EXPERIMENTS

A. Non-Invasive Fetal ECG Database

This study evaluates the performance of the proposed method using real AECG recordings collected from public databases. To date, the 2013 Noninvasive Fetal ECG PhysioNet/Computing in Cardiology Challenge Database(PCDB) is the largest public available dataset, and it is available on PhysioNet [9] (<https://physionet.org/content/challenge-2013/1.0.0/>). Set-A of the PCDB database has a set of 75 one-minute abdominal electrocardiogram records. Each recording consists of four non-invasive abdominal signals, sampled at 1000 Hz, with Fetal QRS reference annotations provided. As indicated in [10], seven AECG records (a33, a38, a47, a52, a54, a71 and a74) are discarded because of inaccurate reference annotations. As suggested in [10], the 68 retained recordings are split into a test set (records a01-a07), validation set (records a08-a13), and a training set (remaining 55 recordings).

B. Evaluation Metrics

In accordance with the ANSI/AAMI guideline [11], three widely used evaluation matrices of R-peak detection tasks, PPV, Recall and F1-score, are utilized to assess fetal R-peak detection performance. These are calculated as follows:

$$PPV = \frac{TP}{TP + FP} \times 100\% \quad (1)$$

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (2)$$

$$F1-score = \frac{2 \times PPV \times Recall}{PPV + Recall} \times 100\% \quad (3)$$

To facilitate fair comparison with previous works, this study also employs a tolerance window of 50ms. This means that if the detected fetal R-peak falls within ± 50 ms of the true fetal R-peak annotation, the detected beat will be classified as a true positive (TP) prediction. The remaining scenarios are categorized as false negative (FN), indicating a missed true annotation, or false positives (FP), signifying an incorrect prediction of the peak.

C. Experimental Method

The proposed model is trained as described in section II-B, using the training dataset described in section III-A. Once trained, the model is then applied to test records a01 to a07, and R-peak locations are predicted as described in section II-C. Evaluation metrics are then determined by comparison of predicted R-peak locations to true locations provided for the dataset.

To evaluate performance, a comparison is conducted with five different methods from the literature which perform direct fetal R-peak detection, without prior cancellation of the maternal ECG signal. These include three machine learning-based approaches: Naive Bayes (NB), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN), which are applied to single-channel input as implemented by and reported in [1]. Two deep learning approaches based on Convolutional Neural Networks (CNN) are also included in the comparison: one utilizing single-channel input (SCI) [1] and the other employing multi-channel input (MCI) [2]. Each method includes pre- and or post-processing as described in [1] and [2]. Metrics for all methods were calculated as described in section III-B.

IV. RESULTS AND DISCUSSION

The effectiveness of the proposed fetal R-peak detection algorithm and the five machine-learning based approaches (see section III-C) are summarised in Table I, which presents the average PPV, Recall and F1-score across the 7 records of the test set. The proposed method shows improve across each metric, with particular improvement in PPV and F1-score.

TABLE I
PERFORMANCE COMPARISON OF PROPOSED UNET-BiLSTM FETAL R-PEAK DETECTION METHOD WITH DIFFERENT MACHINE LEARNING OR DEEP LEARNING APPROACHES USING SET-A OF PCDB DATABASE.

Methods	PPV	Recall	F1-SCORE
NB [1]	56	70	45.13
SVM [1]	68	78	72.66
KNN [1]	70	64	66.87
CNN + SCI [1]	76	78	76.99
Deep CNN + MCI [2]	92.77	89.06	90.88
Proposed Method	95.49	90.85	93.01

Further indications of the performance of UNet-BiLSTM can be seen in Table II, where metrics for each record are shown. Results are compared to those for CNN+MCI [2] which had the next best average performance. Results for test records a01, a02, a03, a04, a05 and a07 show clear improvement using UNet-BiLSTM, all each case having reduced FP and FN predictions. For record a06, both methods has similar PPV, with UNet-BiLSTM scoring a slightly higher score of 88.46%, reflecting a 0.35% improvement compared to [2], translating to fewer false positive predictions. However this was at the expense of a 2.96% lower Recall compared to [2],

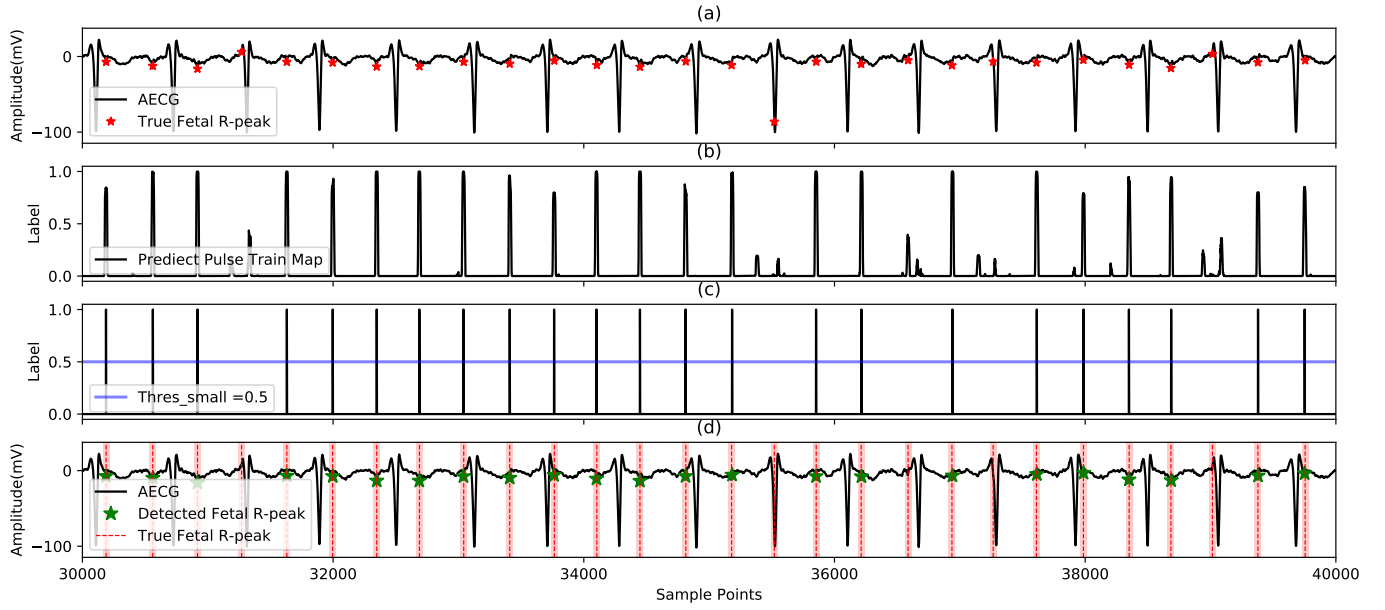


Fig. 3. An example of (a) The Abdominal ECG signal with true fetal R-peak labeled as red star. (b) Pulse train map predicted by proposed method. (c) Fetal R-peak map obtained after post-processing by fetal R-peak detection scheme. (d) Abdominal ECG signal showing final UNet-BiLSTM predicted fetal R-peak locations as green stars. Red lines show true R-peak locations, and tolerance (match window) indicated by red highlight.

TABLE II
COMPARISON OF PERFORMANCE ON EACH TEST RECORD FOR PROPOSED UNET-BiLSTM AND CNN BASED METHOD [2].

RECORD	Deep CNN + MCI [2]			Proposed Method		
	PPV(%)	Recall(%)	F1-score(%)	PPV(%)	Recall(%)	F1-score(%)
a01	99.27	94.44	96.79	99.30	97.24	98.26
a02	81.66	86.79	84.15	94.56	86.88	90.55
a03	92.31	95.24	93.75	96.88	96.88	96.88
a04	100.00	98.44	99.21	100.00	100.00	100.00
a05	99.22	100.00	99.61	100.00	100.00	100.00
a06	88.15	74.84	80.95	88.46	71.88	79.31
a07	88.79	73.64	80.51	89.26	83.08	86.06
Average	92.77	89.06	90.88	95.49	90.85	93.01

TABLE III
COMPARISON OF THE PERFORMANCE OF PROPOSED METHOD FOR DIFFERENT VALUES OF POST-PROCESSING THRESHOLD.

$THRES_{small}$	PPV(%)	Recall(%)	F1-score(%)
0.1	92.04	91.15	91.56
0.3	93.73	91.10	92.35
0.5	95.49	90.85	93.01
0.7	96.92	87.74	91.77
0.9	98.73	82.54	89.22

indicating a higher presence of false negative or missed beat predictions.

Figure 3 visually depicts example incidences of false negative predictions, such as that resulting in poorer performance in record a06. Figure 3(b) shows the predicted pulse train predicted by UNet-BiLSTM model. In most cases, weak

predictions have been made but are then removed as part of the post-processing stage threshold stage, the result of which is shown in Figure 3(c), resulting in a missed prediction (FN). As described in section II-C, this threshold $THRES_{small}$ (set to a fixed value) is applied to the predicted pulse train to eliminate false positive predictions with small magnitudes. The effect of $THRES_{small}$ on resulting performance was evaluated for values ranging from 0.1 to 0.9. Average results for different $THRES_{small}$ values is shown in Table III. The trade off between FP and FN as shown by PPV vs Recall is quickly apparent, with lower thresholds reducing missed beats but increasing fast positive predictions, and higher thresholds increasing missed beats but reducing false positive predictions. $THRES_{small} = 0.5$ was selected (and used for all preceding results) as it provided a trade-off between these competing objectives.

V. CONCLUSION

This study proposes a novel deep learning architecture for direct detection of fetal heartbeats from abdominal electrocardiogram (AECG) signals without the need for maternal electrocardiogram (MECG) cancellation. The architecture combines U-Net structure, bidirectional long short-term memory (Bi-LSTM) and residual paths, using a single-channel AECG signal as input. The experimental results on PhysioNet Cardiology Challenge Database (PCDB) highlight the robust performance of the proposed shallow UNet-BiLSTM model in detecting fetal R-peaks through AECG signals, outperforming other methods applied to the same task.

REFERENCES

- [1] W. Zhong, L. Liao, X. Guo, G. Wang, A deep learning approach for fetal qrs complex detection, *Physiological Measurement* 39 (4) (2018) 045004.
- [2] J. S. Lee, M. Seo, S. W. Kim, M. Choi, Fetal qrs detection based on convolutional neural networks in noninvasive fetal electrocardiogram, in: 2018 4th International Conference on Frontiers of Signal Processing(ICFSP), 2018, pp. 75–78.
- [3] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5–9, 2015, Proceedings, Part III* 18, Springer, 2015, pp. 234–241.
- [4] N. Siddique, S. Paheding, C. P. Elkin, V. Devabhaktuni, U-net and its variants for medical image segmentation: A review of theory and applications, *IEEE Access* 9 (2021) 82031–82057.
- [5] M. U. Zahid, S. Kiranyaz, T. Ince, O. C. Devecioglu, M. E. Chowdhury, A. Khandakar, A. Tahir, M. Gabbouj, Robust r-peak detection in low-quality holter ecgs using 1d convolutional neural network, *IEEE Transactions on Biomedical Engineering* 69 (1) (2021) 119–128.
- [6] R. He, Y. Liu, K. Wang, N. Zhao, Y. Yuan, Q. Li, H. Zhang, Automatic detection of qrs complexes using dual channels based on u-net and bidirectional long short-term memory, *IEEE journal of biomedical and health informatics* 25 (4) (2020) 1052–1061.
- [7] J. Yao, Y. Zhang, C. Dong, A lightweight r peak detection algorithm for noisy ecg signals, in: 2022 IEEE 22nd International Conference on Bioinformatics and Bioengineering (BIBE), 2022, pp. 255–260.
- [8] M. Yu, Z. Huang, Y. Zhu, P. Zhou, J. Zhu, Attention-based residual improved u-net model for continuous blood pressure monitoring by using photoplethysmography signal, *Biomedical Signal Processing and Control* 75 (2022) 103581.
- [9] A. L. Goldberger, L. A. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, H. E. Stanley, Physiobank, physiotoolkit, and physionet: components of a new research resource for complex physiologic signals, *circulation* 101 (23) (2000) pp 215–220.
- [10] J. Behar, J. Oster, G. D. Clifford, Non-invasive FECG extraction from a set of abdominal sensors, *IEEE*, 2013.
- [11] N. Association for the Advancement of Medical Instrumentation, et al., Testing and reporting performance results of cardiac rhythm and segment measurement algorithms, *ANSI/AAMI EC38 1998* (1998) 46.